From Realistic Breast Phantoms to Multimodal AI for Precision Breast Cancer

How can realistic computational representations of the breast and heterogeneous imaging and biological data be combined into robust, clinically useful AI for breast cancer?

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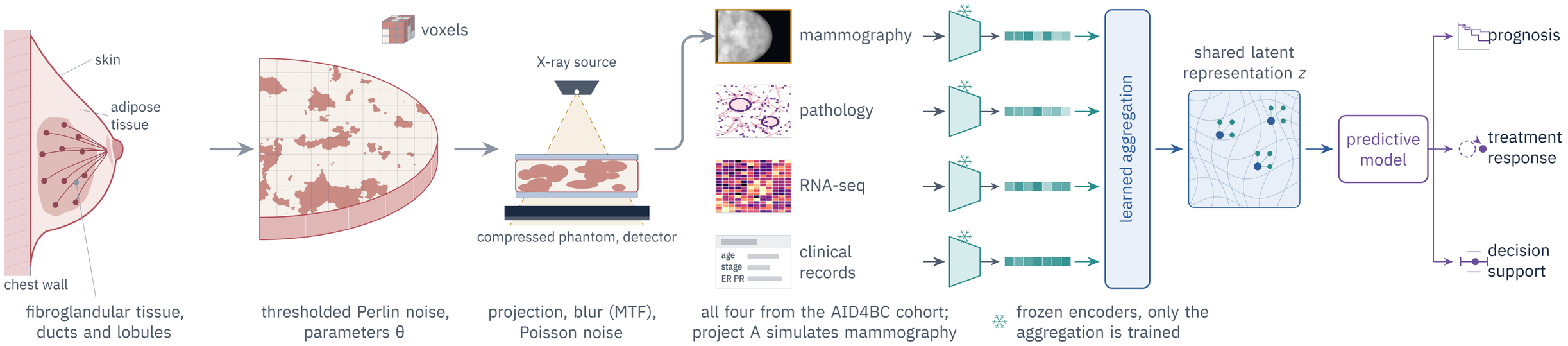
WALLENBERG AI, AUTONOMOUS SYSTEMS AND SOFTWARE PROGRAM

One research pipeline, from the physical breast to clinically relevant predictions

PROJECT A · REALISTIC PHANTOMS, PHYSICS-CONSISTENT IMAGING

PROJECT B · MULTIMODAL REPRESENTATION LEARNING

1 Breast anatomy · 2 3D computational phantom · 3 Imaging physics · 4 Heterogeneous modalities · 5 Modality embeddings · 6 Aggregation and shared latent space · 7 Clinically relevant predictions



Simulated images come with **known ground truth**

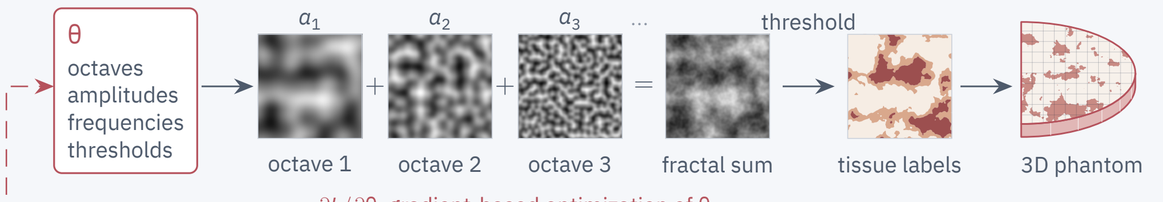
Missing modalities, asynchrony, few samples

Research goal. Prospective validation comes first

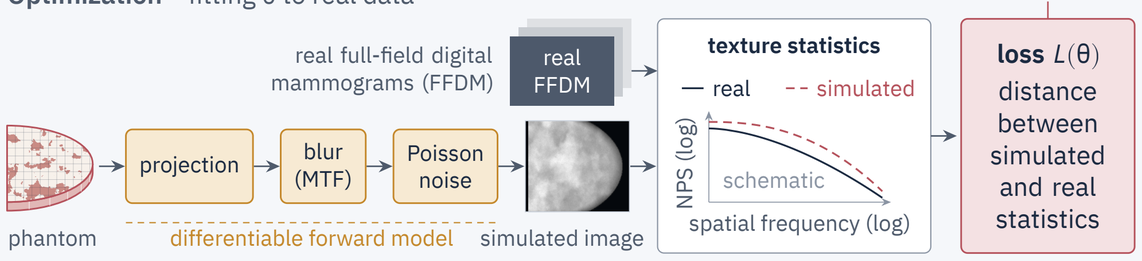
A Realistic breast phantoms, optimized against real mammograms

Virtual clinical trials replace patients with phantoms. A phantom is only as good as its images, so realism must be *measured*, not judged by eye.

Generation parameters θ → 3D tissue structure



Optimization fitting θ to real data



Validation before fitting loop closure, also called parameter recovery



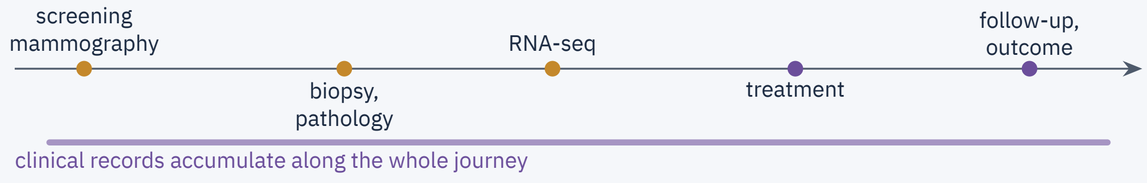
If the method cannot recover parameters it generated itself, a fit to real data cannot be trusted. This self-consistency study is the first milestone.

- **Generation.** Thresholded Perlin-noise octaves (Lund generator; Dustler 2015, 2018; Tomic 2025). θ sets the octave ladder and thresholds.
- **Measurement model.** A differentiable forward model adds projection, MTF blur, and reparameterized Poisson noise; gradients reach θ.
- **Objective.** Match simulated texture to real mammograms (FFDM, MBTST-DM, AIDA Data Hub) on the noise power spectrum, not by eye.
- **Validation first.** No fit to real data before loop closure succeeds; the manuscript is in preparation.

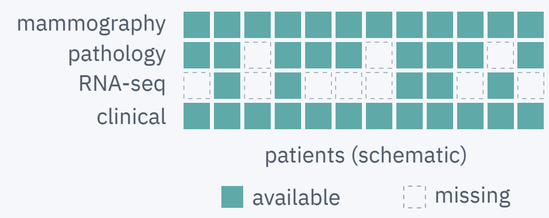
B Multimodal representation learning for precision breast cancer

Mammograms, pathology, RNA-seq, and clinical records arrive at different times and are read in isolation. The goal: *one* shared representation.

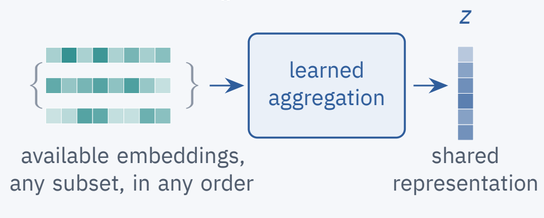
Asynchronous each modality arrives at its own point in the journey



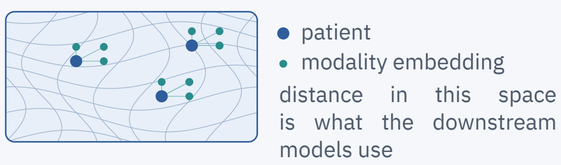
Incomplete few patients carry every modality



for one patient, only the **available** embeddings enter



Shared different modality sets, one space



- **Design principle.** Frozen foundation models encode each modality; effort goes into aggregation, as cohorts are small by machine-learning standards.
- **Representation level.** No raw-data concatenation: a learned aggregation maps the *available* embeddings to a shared representation *z*.
- **Challenges.** Missing modalities, asynchronous acquisition, few labeled samples, calibrated uncertainty.
- **Status.** Aggregation studied on public TCGA data (2026). AID4BC multi-site cohort being assembled; data access pending.

Why the two directions belong together

Phantoms give known ground truth and controllable acquisition physics; cohorts give clinical reality. Simulated data can stress-test multimodal models under missing modalities and acquisition shifts, and quantitative realism keeps the simulation trustworthy. A research vision, not yet tested.

Open research questions

Which image statistics make a phantom realistic *enough* for a given downstream task?

How to aggregate heterogeneous, asynchronous embeddings when few patients carry every modality?

How should uncertainty propagate from acquisition and missing data to a clinical prediction?

Clinical relevance and outlook

The endpoint is decision support for breast cancer care: prognosis and treatment response under calibrated uncertainty, and more equal access to advanced diagnostics. Next are parameter recovery, the MBTST-DM fit, a generative Perlin extension, and aggregation on AID4BC.

TAKE-HOME **Build the breast realistically and validate it quantitatively. Integrate heterogeneous evidence at the representation level. Trustworthy AI for precision breast cancer needs both.**

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